WORKED SOLUTION

# STEP III 2009, Question 2

Pure • Power Series Solutions

**STEP III 2009, Question 2**

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/09-s3-q2

## The key idea

If $y$ is a power series, then so are $y'$, $y''$ and $x^3y$, and the differential equation says that one combined power series is identically zero. A power series that vanishes for every $x$ has every coefficient zero, so the differential equation becomes a list of simple equations linking the $a_n$. The real work is bookkeeping: shift indices so that every series is written in the same power of $x$ before comparing.

## Solution

### Part (i)

Differentiating term by term (we are told all the series converge, so we take this as allowed),

$$\frac{\mathrm{d}}{\mathrm{d}x}(a_n x^n) = n a_n x^{n-1}.$$

The $n = 0$ term $a_0$ is a constant, and its derivative is 0, so the sum starts at $n = 1$:

$$y' = \sum_{n=1}^{\infty} n a_n x^{n-1}.$$

That is the required result. Doing the same again, the $n = 1$ term $a_1$ of $y'$ is constant, so

$$y'' = \sum_{n=2}^{\infty} n(n-1) a_n x^{n-2}.$$

Written out as far as $a_3$:

$$\begin{aligned} y &= a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \cdots, \\ y' &= a_1 + 2a_2 x + 3a_3 x^2 + \cdots, \\ y'' &= 2a_2 + 6a_3 x + \cdots. \end{aligned}$$

**Part (ii)**

We write every term of $xy'' - y' + 4x^3y$ as a series in $x^{n-1}$. First,

$$xy'' = \sum_{n=2}^{\infty} n(n-1)a_n x^{n-1}.$$

The $n = 1$ term of this sum would be $1 \cdot 0 \cdot a_1 = 0$, so we may start it at $n = 1$ too. Then

$$xy'' - y' = \sum_{n=1}^{\infty} [n(n-1) - n]a_n x^{n-1} = \sum_{n=1}^{\infty} n(n-2)a_n x^{n-1}.$$

For the last term, $4x^3y = \sum_{m=0}^{\infty} 4a_m x^{m+3}$, and putting $m = n - 4$ makes the power $x^{n-1}$:

$$4x^3y = \sum_{n=4}^{\infty} 4a_{n-4}\, x^{n-1}.$$

**Ask yourself** Why can we compare coefficients at all?

The equation says $xy'' - y' + 4x^3y = 0$ for every $x$ in the interval where the series converge, so the combined power series is the zero function there. A convergent power series is zero on an interval only if all its coefficients are zero (put $x = 0$ to get the constant term, divide by $x$, and repeat). So the coefficient of each power of $x$ must vanish.

The coefficients are:

- $x^0$ ($n = 1$): $1 \cdot (-1)\, a_1 = 0$, so $a_1 = 0$, as required.
- $x^1$ ($n = 2$): $2 \cdot 0 \cdot a_2 = 0$, which holds automatically, so $a_2$ is free.
- $x^2$ ($n = 3$): $3 \cdot 1 \cdot a_3 = 0$, so $a_3 = 0$.
- $x^{n-1}$ for $n \geq 4$: $n(n-2)a_n + 4a_{n-4} = 0$.

Since $n(n-2) \neq 0$ for $n \geq 4$, we may divide by it, which gives

$$a_n = -\frac{4}{n(n-2)}\, a_{n-4} \qquad (n \geq 4),$$

as required. ($a_0$ is also free, because no equation involves it on its own.)

**The case $a_0 = 1$, $a_2 = 0$.** The recurrence links $a_n$ only to $a_{n-4}$, so the coefficients split into four chains according to $n$ modulo 4. The chains starting at $a_1 = 0$, $a_2 = 0$ and $a_3 = 0$ are zero throughout, so only $a_{4k}$ can be non-zero. Putting $n = 4k$ with $k \geq 1$,

$$a_{4k} = -\frac{4}{4k(4k-2)}\, a_{4k-4} = -\frac{1}{2k(2k-1)}\, a_{4k-4}.$$

Starting from $a_0 = 1$, each step multiplies by $-1$ and divides by the next two factors of a factorial, so

$$a_{4k} = \frac{(-1)^k}{(2k)!}.$$

(Check: $a_4 = -\frac{1}{2}$, $a_8 = -\frac{1}{12}\cdot(-\frac{1}{2}) = \frac{1}{24}$.) Hence

$$y = \sum_{k=0}^{\infty} \frac{(-1)^k x^{4k}}{(2k)!} = \sum_{k=0}^{\infty} \frac{(-1)^k (x^2)^{2k}}{(2k)!} = \cos(x^2),$$

by the Maclaurin series $\cos u = \sum_k (-1)^k u^{2k}/(2k)!$ with $u = x^2$. This is the required result.

**The case $a_0 = 0$, $a_2 = 1$.** Now the chains starting at $a_0$, $a_1$ and $a_3$ are all zero, and only $a_{4k+2}$ survive. Putting $n = 4k+2$ with $k \geq 1$,

$$a_{4k+2} = -\frac{4}{(4k+2)\,4k}\, a_{4k-2} = -\frac{1}{(2k+1)\,2k}\, a_{4k-2},$$

and from $a_2 = 1$ we get $a_{4k+2} = (-1)^k/(2k+1)!$. Therefore

$$y = \sum_{k=0}^{\infty} \frac{(-1)^k x^{4k+2}}{(2k+1)!} = \sum_{k=0}^{\infty} \frac{(-1)^k (x^2)^{2k+1}}{(2k+1)!} = \sin(x^2).$$

## Checks

Substitute directly. For $y = \sin(x^2)$, $y' = 2x\cos(x^2)$ and $y'' = 2\cos(x^2) - 4x^2\sin(x^2)$, so

$$xy'' - y' + 4x^3 y = 2x\cos(x^2) - 4x^3\sin(x^2) - 2x\cos(x^2) + 4x^3\sin(x^2) = 0.$$

The same works for $\cos(x^2)$. Numerically, the first 40 terms of the cosine series at $x = 1.2$ give $0.1304237\ldots$, matching $\cos(1.44)$. Also, both answers make sense at $x = 0$: $\cos 0 = 1 = a_0$, and $\sin(x^2) \approx x^2$ near 0, so $a_2 = 1$.

## Common slips

- Comparing coefficients before shifting indices, so that $a_n$ in one series is matched with $a_n$ in another even though they multiply different powers of $x$.
- Forgetting that $a_3 = 0$, which is needed to kill the chain $a_3, a_7, a_{11}, \ldots$.
- Dividing by $n(n-2)$ without noting that it is non-zero for $n \geq 4$. At $n = 2$ it is zero, which is exactly why $a_2$ is free.

## Beyond STEP

This is the method of Frobenius in its simplest form. The point $x = 0$ is a *regular singular point* of this equation, because the coefficient of $y''$ vanishes there, and the general theory explains why one ordinary power series succeeds here when it sometimes fails (the indicial roots, 0 and 2, differ by an integer). The same method gives Bessel and Legendre

functions. Justifying term-by-term differentiation needs uniform convergence, which is first-year analysis. For both ideas, see Boyce and DiPrima, *Elementary Differential Equations*, and Abbott, *Understanding Analysis*.

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